

Connecting Specific and Shannon Entropies via the Gravitational Field

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Abstract. Three measures of the entropy of the observable universe are computed in a single accounting, with every entropy expressed in units of k_B . Counting the non-dark particle content, with the statistical factors per particle stated in advance, gives $S_{\text{bary}} \approx 7 \times 10^{89} k_B$. The Shannon measure, the ratio of the universe’s inertial mass-energy to its Planck energy $\hbar H_0$, gives $S_{MI} \approx 4 \times 10^{120} k_B$, which is equivalently the maximal causal angular momentum McR_H in units of $\hbar$. The gap between the two is therefore some thirty-one orders of magnitude, and the candidate that closes it is the entropy of the gravitational field itself: an Ellis-type effective thermodynamics with a radiation-like equation of state and a de Sitter temperature $T_g = \hbar H_0 / 2\pi k_B$ yields $S_{\text{grav}} = \frac{4}{3} Mc^2 / (k_B T_g) \approx 3 \times 10^{122} k_B$, within two orders of the holographic bound $\pi R_H^2 / \ell_P^2 \approx 2 \times 10^{122} k_B$ and exceeding S_{MI} by the fixed factor $16\pi^2/3$. All three ceilings are $\mathcal{O}(1)$ – $\mathcal{O}(10)$ multiples of the reciprocal of the single Buckingham invariant $\Pi_0 = \hbar G H_0^2 / c^5 \approx 10^{-122}$, so the cosmological hierarchy and the universe’s entropy ceiling are one number read from opposite sides. Fractional responses to CMB-scale perturbations, dS/S , are computed for every measure and lie between 4×10^{-8} and 4×10^{-5} , all set by the observed anisotropy $dT/T \sim 4 \times 10^{-6}$; the fractional growth rate of the Shannon measure is $\dot{S}/S = -\dot{H}/H \simeq H_0$, so entropy growth slows as $1/t$ and approaches a ceiling $Mc^2/(hH_\infty)$ in the dark-energy era.

1 Three measures of Entropy

The entropy of the universe can be estimated by counting what it contains, by weighing it, or by asking its gravitational field. The first, S_{bary} , sums the entropy of the photons, neutrinos and baryons of the non-dark sector. The second, S_{MI} , is the Shannon information content [15, 16], the dimensionless ratio of the inertial mass-energy Mc^2 to the quantum of energy the cosmos itself defines, $\hbar H_0$. The third, S_{grav} , is the entropy Ellis and collaborators [25] assign to the free gravitational field of an inhomogeneous universe.

The plan of this paper is to compute all three in one consistent unit, the Boltzmann constant, so that they can be honestly compared; to compute for each the fractional response dS/S to a perturbation at the scale of the observed CMB anisotropy; and to let the comparison, rather than any single number, carry the argument. Throughout, $H_0 \approx 2.2 \times 10^{-18} \text{ s}^{-1}$, the Hubble radius is $R_H = c/H_0 \approx 1.4 \times 10^{26} \text{ m}$, the comoving radius of the observable universe is $R_{\text{obs}} \approx 4.4 \times 10^{26} \text{ m}$ with volume $V \approx 4 \times 10^{80} \text{ m}^3$, and the inertial mass is taken as $M \approx 10^{53} \text{ kg}$. The two radii are distinct and are used for distinct purposes: V bounds the particle inventory, R_H sets causal and horizon quantities.

2 The non-dark inventory

The statistical mechanics enters through one number per species, the entropy carried per particle, and these factors are stated first because everything downstream is a multiplication by them. For a boson gas in equilibrium the entropy per photon is a pure number,

$$s_\gamma = \frac{S_\gamma}{N_\gamma k_B} = \frac{2\pi^4}{45 \zeta(3)} \approx 3.60, \quad (1)$$

the ratio of the entropy density $\frac{4}{3}aT^3$ to the number density bT^3 , with $\zeta(3)$ arising from the Bose–Einstein integral (Appendix A). For the relativistic Fermi–Dirac neutrino background the corresponding factor is $s_\nu \approx 3$ [9, 8], reduced from the photon value by the 7/8 family of fermionic integrals. For the non-relativistic, distinguishable baryons the classical count gives $s_b \approx \ln N_b \approx 184$ per particle. Number densities are fixed by the CMB temperature $T_\gamma = 2.725 \text{ K}$, the decoupling-cooled neutrino temperature $T_\nu = (4/11)^{1/3} T_\gamma \approx 1.95 \text{ K}$ [14, 7], and the baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$. Table 1 is then the whole of the non-dark accounting.

Two remarks tie the table to familiar quantities. The specific entropy, the photon count per baryon, is

$$S_{\gamma b} = \frac{n_\gamma}{n_b} = \frac{1}{\eta} \approx 1.7 \times 10^9, \quad (2)$$

Table 1: The non-dark inventory of the observable universe ($V = 4 \times 10^{80} \text{ m}^3$). Internal energies are thermal; the baryon rest-mass energy $M_b c^2 \approx 1.5 \times 10^{70} \text{ J}$ is excluded from the thermodynamic accounting by the baryon chemical potential (Sec. 3).

Species	Statistics	$n [\text{m}^{-3}]$	N	s/k_B per particle	$S [k_B]$	$U_{\text{th}} [\text{J}]$
photons	Bose–Einstein	4.1×10^8	1.6×10^{89}	3.60	5.9×10^{89}	1.7×10^{67}
neutrinos	Fermi–Dirac	1.1×10^8	4.5×10^{88}	≈ 3	1.3×10^{89}	3×10^{66}
baryons	classical	2.5×10^{-1}	9.9×10^{79}	$\ln N_b \approx 184$	1.8×10^{82}	5.5×10^{57}
total					$S_{\text{bary}} \approx 7.3 \times 10^{89}$	$U \approx 2.0 \times 10^{67}$

in the sense of Prigogine [1, 2], and the entropy per baryon that follows from the table, $S_{\text{bary}}/N_b \approx 7 \times 10^9 k_B$, is $s_\gamma S_\gamma$ to within the neutrino contribution, as it must be. Second, the photon internal energy obeys the black-body identity $T_\gamma S_\gamma = \frac{4}{3} U_\gamma$, with $U_\gamma = a T_\gamma^4 V = 1.7 \times 10^{67} \text{ J}$ and $T_\gamma S_\gamma k_B = 2.2 \times 10^{67} \text{ J}$; the internal energies appear in the table because the fluctuation analysis of Sec. 4 differentiates them, and for no other reason, the entropies themselves being the primary objects of the paper.

3 The thermodynamic relation, and Chemical Potential

The accounting rests on the second law in the form

$$k_B T dS = dU + p dV + \mu dN, \quad (3)$$

with S in units of k_B (nats), and with the radiation equation of state $p = \rho/3$ following from the isotropy of the energy-momentum tensor of a perfect photon fluid. Setting $\mu = 0$ while subsequently computing particle-number changes dN looks like a contradiction, and resolving it is worth a short section because the resolution is exact rather than approximate.

For photons, express $U = a T^4 V$ through $N = b T^3 V$ to obtain $U = a b^{-4/3} N^{4/3} V^{-1/3}$, whence the energy cost of one more photon at fixed volume is

$$\begin{aligned} \left(\frac{\partial U}{\partial N} \right)_V &= \frac{4}{3} a b^{-4/3} N^{1/3} V^{-1/3} \\ &= 1.36 \times 10^{-22} \text{ J} = 3.60 k_B T_\gamma. \end{aligned} \quad (4)$$

The number is no accident: it equals T_γ times the entropy per photon of Eq. (1). A photon added at fixed volume brings energy $3.6 k_B T$ and entropy $3.6 k_B$, so the full chemical potential vanishes identically,

$$\mu_\gamma = \left(\frac{\partial U}{\partial N} \right)_V - T \left(\frac{\partial S}{\partial N} \right)_V = 3.6 k_B T - 3.6 k_B T = 0. \quad (5)$$

Photon number is therefore a dependent variable, free to change at zero thermodynamic cost, and Eq. (3) with $\mu_\gamma = 0$ remains exact while dN_γ is computed from dS_γ as bookkeeping. The same holds for the neutrino background after decoupling. For baryons the opposite limit applies: baryon number is conserved and the chemical potential is dominated by the rest energy, $\mu_b \approx m_b c^2$, which is precisely why the rest-mass energy $M_b c^2 \approx 1.5 \times 10^{70} \text{ J}$, though eight hundred times the photon energy, does not enter the entropy budget: the $\mu_b dN_b$ term cancels it from $k_B T dS$, leaving only the thermal $U_b = \frac{3}{2} N_b k_B T_b = 5.5 \times 10^{57} \text{ J}$ to participate.

4 Fluctuations at the CMB scale

The observed CMB anisotropy, $dT \approx 10^{-5} \text{ K}$ on $T_\gamma = 2.725 \text{ K}$ [10, 11], so $dT/T \approx 3.7 \times 10^{-6}$, is adopted as the universal perturbation scale. For radiation species $S \propto T^3$ at fixed volume, so the fractional response is immediate,

$$\frac{dS}{S} = 3 \frac{dT}{T}, \quad (6)$$

giving $dS_\gamma/S_\gamma = 1.1 \times 10^{-5}$ and, at the cooler neutrino temperature, $dS_\nu/S_\nu = 1.5 \times 10^{-5}$. For baryons, taking the density perturbation $dn_b \approx 10^{-5} \text{ m}^{-3}$ ($dn_b/n_b = 4 \times 10^{-5}$), the response is $dS_b = (\ln N_b + 1) dN_b$ with $dN_b = V dn_b$. Table 2 collects the results, together with the particle-number changes $dN = dS/s$ that the vanishing chemical potential licenses.

The photon-number change $dN_\gamma/N_\gamma \approx 10^{-5}$ should be set against the purely statistical expectation $dN_{\text{stat}}/N_\gamma \sim N_\gamma^{-1/2} \approx 2.5 \times 10^{-45}$: the thermodynamic response exceeds the Poisson floor by forty orders of magnitude, which is the quantitative sense in which the CMB fluctuations are dynamical rather than statistical. The physical driver is the photon-multiplier action of structure: gravitational clumping concentrates mass

Table 2: Entropy and number responses to a CMB-scale perturbation. Because $S = sN$ species-wise, $dN/N = dS/S$ throughout.

	$dS [k_B]$	dS/S	dN
photons	6.5×10^{84}	1.1×10^{-5}	1.8×10^{84}
neutrinos	2.1×10^{84}	1.5×10^{-5}	6.9×10^{83}
baryons	7.4×10^{77}	4.1×10^{-5}	4.0×10^{75}
grav. field	1.4×10^{115}	4.4×10^{-8}	—

into low-entropy configurations, and dust and stars re-radiate absorbed starlight as many infrared photons at 10–100 K, feeding the cosmic infrared background, conserving energy while multiplying carriers, and pushing the photon count, and with it S_γ , upward at zero chemical cost.

5 The Shannon measure, and universe’s angular momentum

The Shannon information content of the universe is the dimensionless ratio

$$S_{MI} = \frac{Mc^2}{hH_0} \approx 6.2 \times 10^{120} \text{ bits}, \quad (7)$$

$$S = S_{MI} k_B \ln 2 \approx 4.3 \times 10^{120} k_B,$$

the mass-energy counted in quanta of the smallest frequency the universe defines, H_0 . Since h carries the dimension of angular momentum, Eq. (7) has a mechanical reading,

$$S_{MI} = \frac{L_{\max}}{h}, \quad L_{\max} = McR_H \approx 3.9 \times 10^{121} \hbar, \quad (8)$$

the angular momentum of the mass of the universe transported at the causal limit around the Hubble radius; S_{MI} is the maximal causal angular momentum in units of h , a ceiling rather than a measurement.

What of the actual angular momentum? For scale, a thousand children’s tops at 10 rad s^{-1} carry $\sim 10^{34} \hbar$, the Solar System $3 \times 10^{77} \hbar$, the Milky Way $2 \times 10^{101} \hbar$. If the spin directions of the $N \sim 2 \times 10^{12}$ Milky-Way-like galaxies are random, as the Zooniverse survey found no reason to doubt [18, 3], the expected residual is

$$\Delta L_U \sim \sqrt{N} L_{\text{MW}} \approx 2.8 \times 10^{107} \hbar, \quad (9)$$

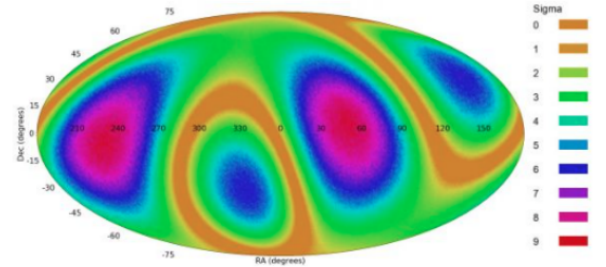
a population-level fractional handedness $\Delta L_U/(NL_{\text{MW}}) = N^{-1/2} \approx 7 \times 10^{-7}$, which is the $\sim 10^{-6}$ anticipated on Landau’s fluctuation grounds and is consistent both with the null

Zooniverse result and with the marginal dipole drift Shamir reports at higher redshift [24], the CMB cold spot of Fig. 1 remaining a suggestive but unproven axis [19]. The resolution of the comparison is then this: if the universe carries only the statistical residual, treating it as rigid rotation of a sphere of mass M and radius R_H with $I = \frac{2}{5}MR_H^2$ implies

$$v = \frac{5}{2} \frac{\Delta L_U}{MR_H} \approx 5 \times 10^{-6} \text{ m s}^{-1}, \quad (10)$$

$$a = \frac{v^2}{R_H} \approx 2 \times 10^{-37} \text{ m s}^{-2},$$

a rotation utterly beneath observation, so the universe can own its random residual, $\Delta L_U/L_{\max} \approx 7 \times 10^{-15}$, without any detectable global spin, and the Shannon ceiling $S_{MI} = L_{\max}/h$ stands regardless of the sign or size of the net handedness. The monopole and dipole modes at the far left of the CMB power spectrum of Fig. 2 are the early-universe artefacts against which any such axis would have to be sought [22].



Bias in galaxy spin directions on a cosmic scale. Credit: Lior Shamir

Figure 1: The great void of the CMB as a candidate axis of rotation.

Comparing Eq. (7) with Table 1 now gives the gap in honest units:

$$\frac{S}{S_{\text{bary}}} \approx \frac{4.3 \times 10^{120}}{7.3 \times 10^{89}} \approx 6 \times 10^{30}, \quad (11)$$

some thirty-one orders of magnitude. Counting the non-dark particles recovers a vanishing fraction of the entropy the universe’s inertia is capable of encoding, and the deficit must live in a sector the particle count cannot see.

6 Gravitational field entropy

The candidate is the gravitational field itself. Ellis and collaborators [25] construct an effective thermodynamics for the free field, sourced by the

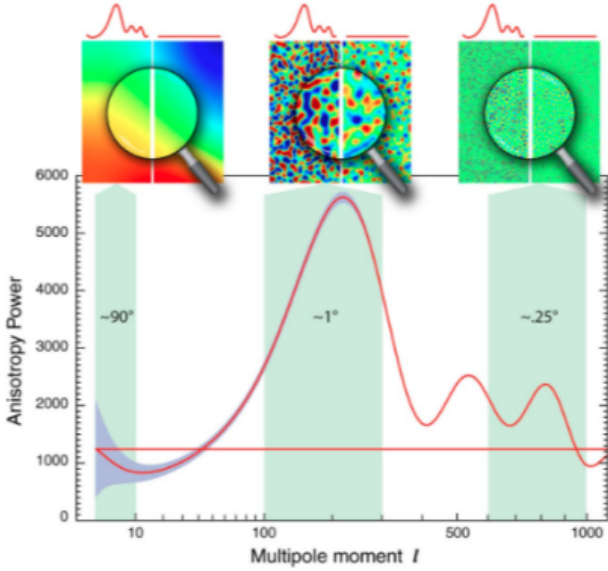


Figure 2: Monopole ($\ell = 1$) and dipole ($\ell = 2$) moments are early-universe CMBR modes.

Bel–Robinson super-energy in place of the ordinary energy-momentum tensor of a perfect fluid, and propose

$$T_g dS_{\text{grav}} = dU_{\text{grav}} + p_{\text{grav}} dV, \quad p_{\text{grav}} = \frac{1}{3} \rho_{\text{grav}}, \quad (12)$$

a radiation-like equation of state for a field whose entropy grows as inhomogeneity grows. Within a marginally bound Lemaître–Tolman–Bondi model, $ds^2 = -dt^2 + (R')^2 dr^2 + R^2 d\Omega^2$ with $E(r) = 0$, the areal radius obeys $\dot{R}^2 = 2GM(r)/R$, integrating to $R \propto (\kappa \rho r^3)^{1/3} t^{2/3}$, so that the dimensionless gradient evaluates at the Hubble scale to

$$R' = \beta (\kappa \rho)^{1/3} t^{2/3} R_H^{1/3} \approx 4 \times 10^{-3}, \quad (13)$$

with $\beta = 2^{1/3} 3^{-2/3}$, and the expansion-rate combination $|\dot{R}'/R'|$ evaluates to a few $\times 10^{-22} \text{ s}^{-1}$, of order H_0 diluted by the same inhomogeneity factors. The temperature assigned to the field is the horizon (Unruh–de Sitter) temperature of that rate,

$$T_g = \frac{\hbar}{2\pi k_B} \left| \frac{\dot{R}'}{R'} \right| \sim \frac{\hbar H_0}{2\pi k_B} \approx 2.7 \times 10^{-30} \text{ K}, \quad (14)$$

the factor $\hbar/k_B$ being required for the dimensions and supplying the quantum of the construction. Taking $\rho_{\text{grav}} c^2 V \approx Mc^2$ for the marginally inhomogeneous field and integrating Eq. (12) at constant ρ_{grav} and T_g ,

$$S_{\text{grav}} = \frac{4}{3} \frac{Mc^2}{k_B T_g} = \frac{8\pi}{3} \frac{Mc^2}{\hbar H_0} \approx 3.2 \times 10^{122} k_B. \quad (15)$$

Three comparisons situate the number. It exceeds the Shannon measure by a fixed, parameter-free ratio,

$$\frac{S_{\text{grav}}}{S_{MI} k_B} = \frac{16\pi^2}{3} \approx 53, \quad (16)$$

because both are $Mc^2/(\hbar H_0)$ dressed in different $\mathcal{O}(10)$ constants: once the field is granted a radiation-like equation of state, so that $T_g S_{\text{grav}} = \frac{4}{3} U_{\text{grav}}$ exactly as $T_\gamma S_\gamma = \frac{4}{3} U_\gamma$ for the photons of Sec. 2, and a temperature $T_g \propto \hbar H_0/k_B$, the coincidence with S_{MI} is an identity rather than a numerical accident, and the physical content of the claim lives entirely in those two choices. It sits within two orders of the holographic bound,

$$S_{\text{holo}} = \frac{\pi R_H^2}{\ell_P^2} \approx 2.2 \times 10^{122} k_B, \quad (17)$$

as any near-saturating assignment must. And it dwarfs the particle inventory by the thirty-plus orders the gap demanded: on this accounting the entropy of the universe is overwhelmingly gravitational, the matter sector a correction in the ninetieth decade, which is the quantitative face of the Penrose intuition that the low-entropy initial condition was a gravitational smoothness whose spending is structure formation.

The fractional response completes Table 2. A CMB-scale perturbation $dr = R_H dT/T \approx 5 \times 10^{20} \text{ m}$ (about 0.15 Mpc) advances the areal radius by $dR_H = R' dr \approx 2 \times 10^{18} \text{ m}$, so

$$\frac{dS_{\text{grav}}}{S_{\text{grav}}} = 3 \frac{dR_H}{R_H} = 3R' \frac{dT}{T} \approx 4.4 \times 10^{-8}, \quad (18)$$

suppressed below the matter-sector responses by the gradient $R' \approx 4 \times 10^{-3}$; in absolute terms $dS_{\text{grav}} \approx 1.4 \times 10^{115} k_B$, twenty-five orders beyond the photon response, so the gravitational field dominates the fluctuation budget as thoroughly as it dominates the total.

7 Invariant, 10^{122} and units

The recurrence of 10^{122} across Eqs. (15)–(17) is neither accident nor independent evidence, and this section traces it to a single dimensionless number, following the Buckingham analysis of [26]. The constants c , $\hbar$ and G have dimension vectors spanning the mechanical space (M, L, T) , their dimensional matrix having determinant -2 and full rank, which is the linear-algebraic licence for the natural-unit choice $c = \hbar = G = 1$; adjoining

the Hubble parameter, of dimension T^{-1} , leaves the rank at three, so by the Π -theorem exactly one dimensionless combination exists,

$$\Pi_0 = \frac{\hbar G H_0^2}{c^5} = (H_0 t_P)^2 \approx 1.4 \times 10^{-122}, \quad (19)$$

a hierarchy that no change of basis can remove, only relocate: in Planck units it appears as $H_0^2 \sim 10^{-122}$, in a Hubble-adapted basis $c = \hbar = H_0 = 1$ as $G \sim 10^{-122}$, and through the de Sitter relation $H^2 = \Lambda c^2/3$ as $\Lambda \ell_P^2 \sim 10^{-122}$ [26].

The bridge from this invariant to the entropies of this paper is the critical density. The unique combination of H_0 and G with the dimension of density is $\rho_c = 3H_0^2/8\pi G$, and the mass it assigns to the Hubble volume,

$$M = \rho_c \cdot \frac{4}{3}\pi R_H^3 = \frac{c^3}{2GH_0} \approx 9.2 \times 10^{52} \text{ kg}, \quad (20)$$

is the $M \approx 10^{53}$ kg adopted throughout. With this M , the ratio $Mc^2/(\hbar H_0) = c^5/(2\hbar GH_0^2) = \frac{1}{2}\Pi_0^{-1}$, and writing

$$N \equiv \Pi_0^{-1} = \left(\frac{R_H}{\ell_P}\right)^2 \approx 7.2 \times 10^{121}, \quad (21)$$

every ceiling in this paper collapses to an $\mathcal{O}(1)$ – $\mathcal{O}(10)$ multiple of the one number:

$$S_{MI} = \frac{N}{4\pi} \text{ bits}, \quad (22)$$

$$S_{\text{grav}} = \frac{4\pi}{3} N k_B, \quad S_{\text{holo}} = \pi N k_B,$$

and the Gibbons–Hawking entropy of the de Sitter horizon, $S_\Lambda = 3\pi k_B/(\Lambda \ell_P^2) = (\pi/\Omega_\Lambda)N k_B$, joins them. The conclusion is a reciprocity: the 10^{-122} of the cosmological hierarchy and the 10^{122} of the universe’s entropy ceiling are the same Buckingham invariant read from opposite sides, $S \sim \pi k_B/\Pi_0$ against $\Lambda \ell_P^2 = 3\Omega_\Lambda \Pi_0$, so that the cosmological constant problem and the question of why the universe can hold so much information are one question in two dimensional bases. A numerical curiosity noted playfully in [26] places the inverse seventh power of the Ramanujan constant, $e^{-7\pi\sqrt{163}} \approx 1.16 \times 10^{-122}$, at the same magnitude, both 7 and 163 being Heegner numbers, though no mechanism is implied!

A closing word on units, since the paper’s accounting has depended on it. The basis $c = \hbar = G = 1$ absorbs the mechanical dimensions entirely, and what remains outside it is temperature: Θ is a fourth base dimension, and k_B is its lone carrier,

exactly as the temperature row of a dimensional matrix pivots on k_B alone. One may therefore legitimately complete the basis with $k_B = 1$, rendering every entropy a pure count; this paper instead keeps k_B explicit as the visible dictionary between the thermodynamic description in JK^{-1} and the informational description in nats and bits, precisely the translation on which the comparison of Secs. 5 and 6 turns, and the invariant Π_0 , being dimensionless, is indifferent to the choice.

8 Entropy evolution, and the three measures sewn together

The Hubble parameter evolves as $H^2 = H_0^2 [\Omega_m(R_0/R)^3 + \Omega_r(R_0/R)^4 + \Omega_\Lambda]$ [21], falling as t^{-1} through the radiation and matter eras and stabilising at $H_\infty = H_0\sqrt{\Omega_\Lambda}$ in the dark-energy era. Both ceiling measures inherit their evolution from H alone: with $k_B T = \hbar H$ as the universe’s own quantum,

$$S_{MI} \propto \frac{1}{H}, \quad \frac{\dot{S}_{MI}}{S_{MI}} = -\frac{\dot{H}}{H} \simeq H \sim \frac{1}{t}, \quad (23)$$

so the fractional growth rate of the universe’s information capacity is today $\approx H_0 = 2.2 \times 10^{-18} \text{ s}^{-1}$, an e -fold per Hubble time, slowing as $1/t$ and freezing at the constant $Mc^2/(\hbar H_\infty)$ once dark energy dominates. The same is true of S_{grav} and S_{holo} , which are the same function of H up to their fixed constants. The picture that sews the three measures together is then a hierarchy with dynamics: the particle entropy $S_{\text{bary}} \approx 7 \times 10^{89} k_B$ grows slowly through photon multiplication at rates set by the 10^{-5} fluctuation scale of Table 2; the gravitational entropy, thirty orders larger, grows as structure deepens, its ceiling S_{holo} receding ever more slowly as H settles; and the Shannon measure names the capacity toward which both climb, reached asymptotically through the inexorably smaller fractional increments of Eq. (23). The universe approaches its own information content from below, gravitationally, at a rate that is today one part in 10^{18} per second and falling.

A Bose–Einstein integral & $\zeta(3)$

The number density of a photon gas integrates the Bose–Einstein distribution over states, $n_\gamma = \int_0^\infty g(\epsilon) (e^{\epsilon/k_B T} - 1)^{-1} d\epsilon$ with $g(\epsilon) = 8\pi\epsilon^2/(hc)^3$;

substituting $x = \epsilon/k_B T$,

$$n_\gamma = \frac{8\pi(k_B T)^3}{(hc)^3} \int_0^\infty \frac{x^2}{e^x - 1} dx = \frac{16\pi\zeta(3)(k_B T)^3}{(hc)^3}, \quad (24)$$

the integral evaluating by geometric expansion, $\int_0^\infty x^2 e^{-nx} dx = \Gamma(3)/n^3 = 2/n^3$, summed to $2\zeta(3)$. The Fermi–Dirac integral gives $\frac{3}{2}\zeta(3)$ *mutatis mutandis*, whence $n_\nu = \frac{3}{11} n_\gamma$ per the decoupling temperature ratio, and the entropy-per-particle factors of Sec. 2 follow as ratios of the corresponding $\zeta(4)$ -type energy integrals to these number integrals.

B Photon chemical potential vanishes identically

Expressing $U = aT^4V$ through $N = bT^3V$ gives $U = a b^{-4/3} N^{4/3} V^{-1/3}$ and hence

$$\begin{aligned} \left(\frac{\partial U}{\partial N}\right)_V &= \frac{4}{3} \frac{U}{N} = \frac{4}{3} \times 2.70 k_B T \\ &= 3.60 k_B T = 1.36 \times 10^{-22} \text{ J}, \end{aligned} \quad (25)$$

the mean photon energy being $2.70 k_B T$ and the factor $\frac{4}{3}$ the usual radiation dressing. Since the entropy per photon is the same number 3.60 in units of k_B , Eq. (1), the Gibbs combination vanishes exactly, $\mu_\gamma = (\partial U/\partial N)_V - T(\partial S/\partial N)_V = 0$: the energy of an added photon is wholly entropic, which is the precise licence for dropping μdN from Eq. (3) while continuing to track dN .

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